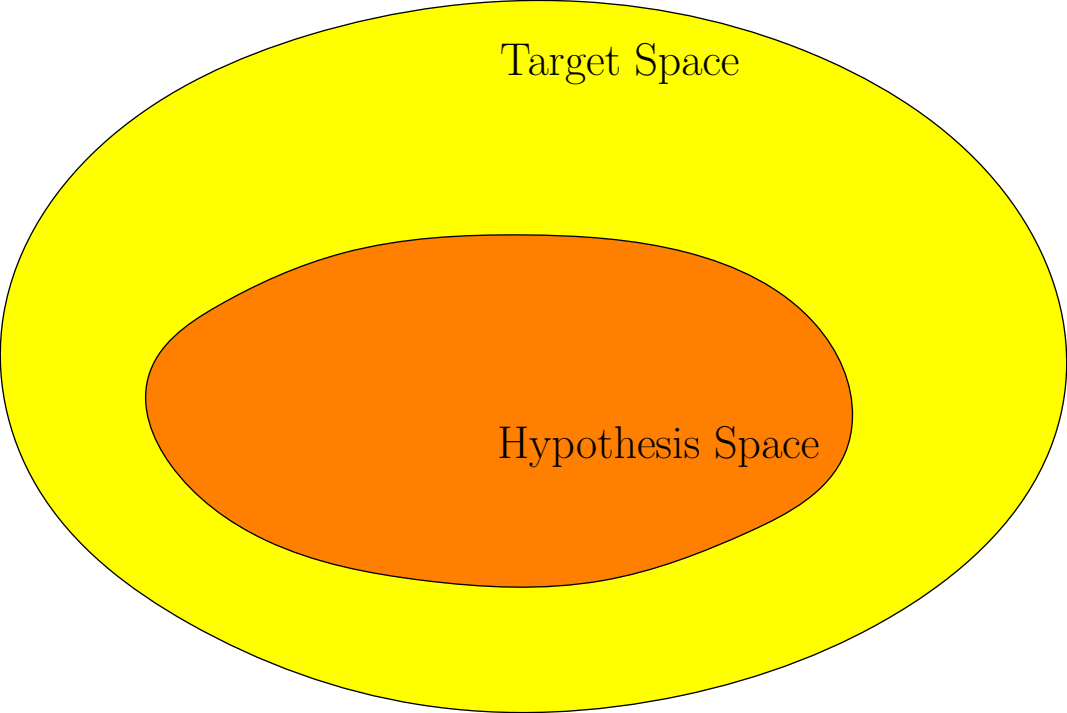
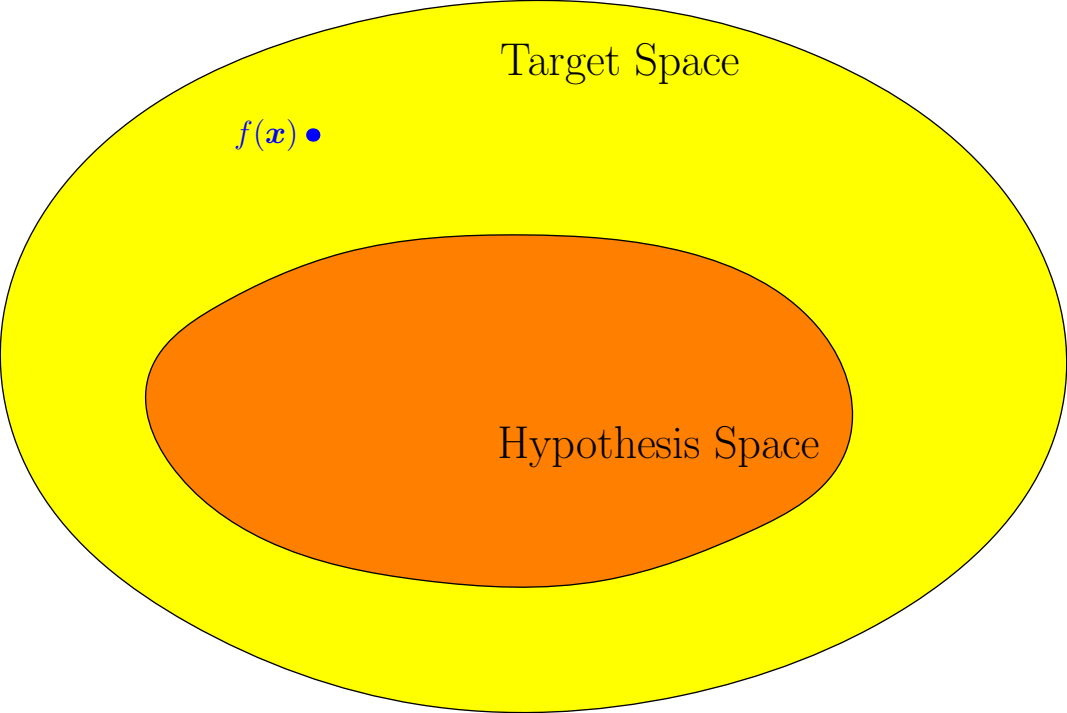
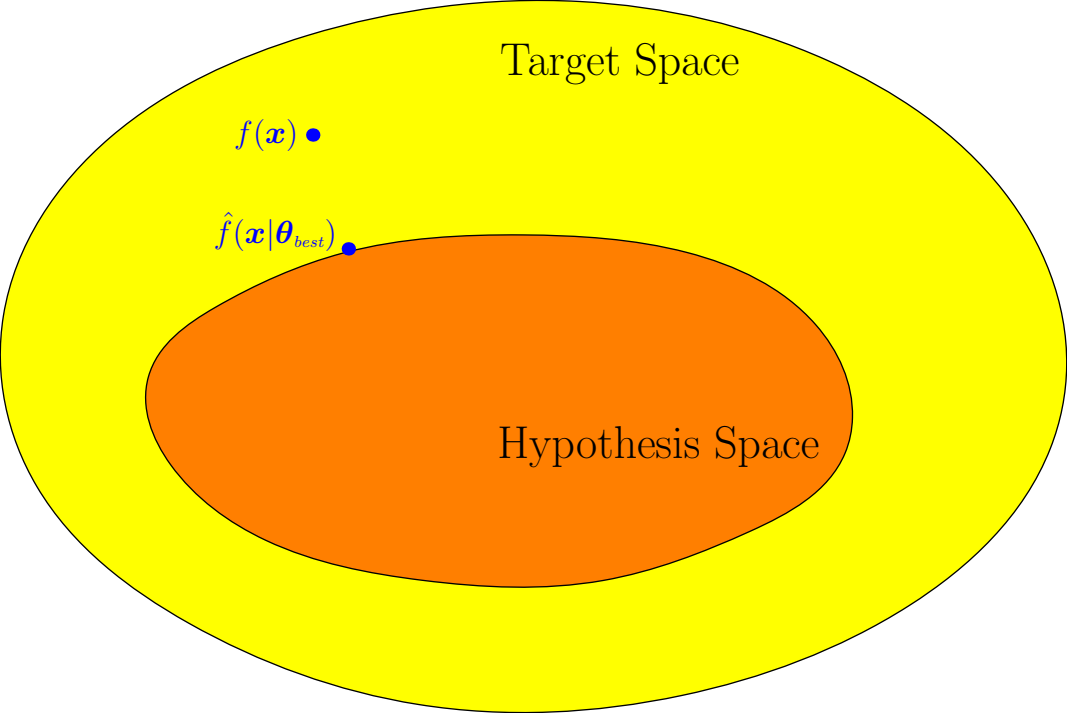
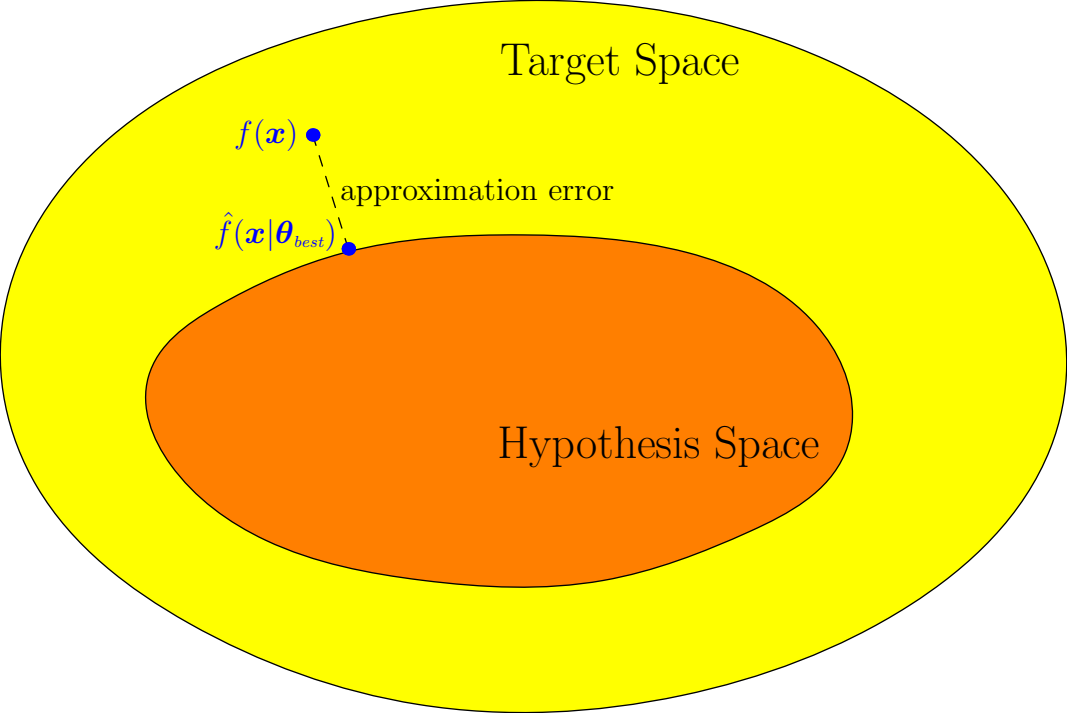
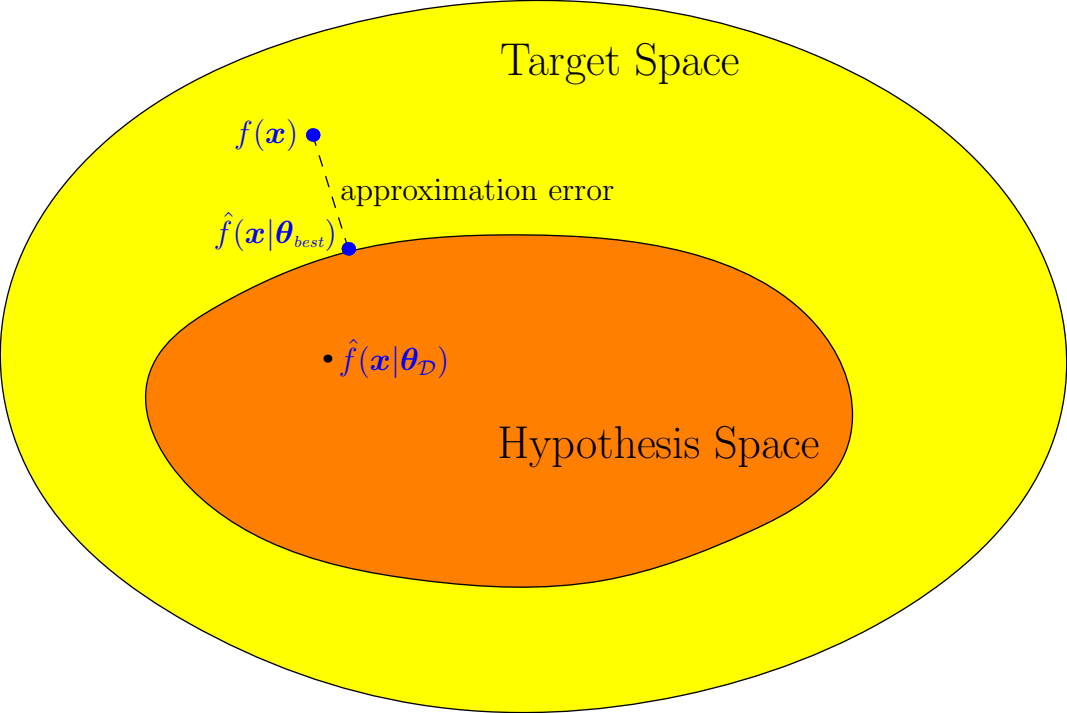
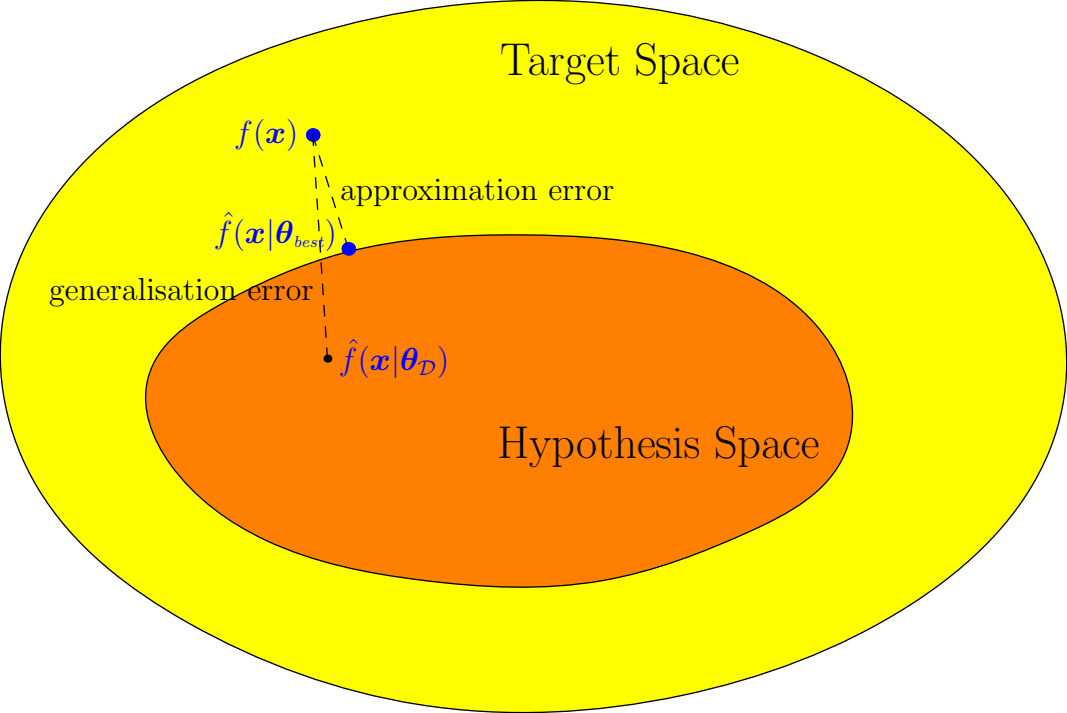

Target Space
Hypothesis Space

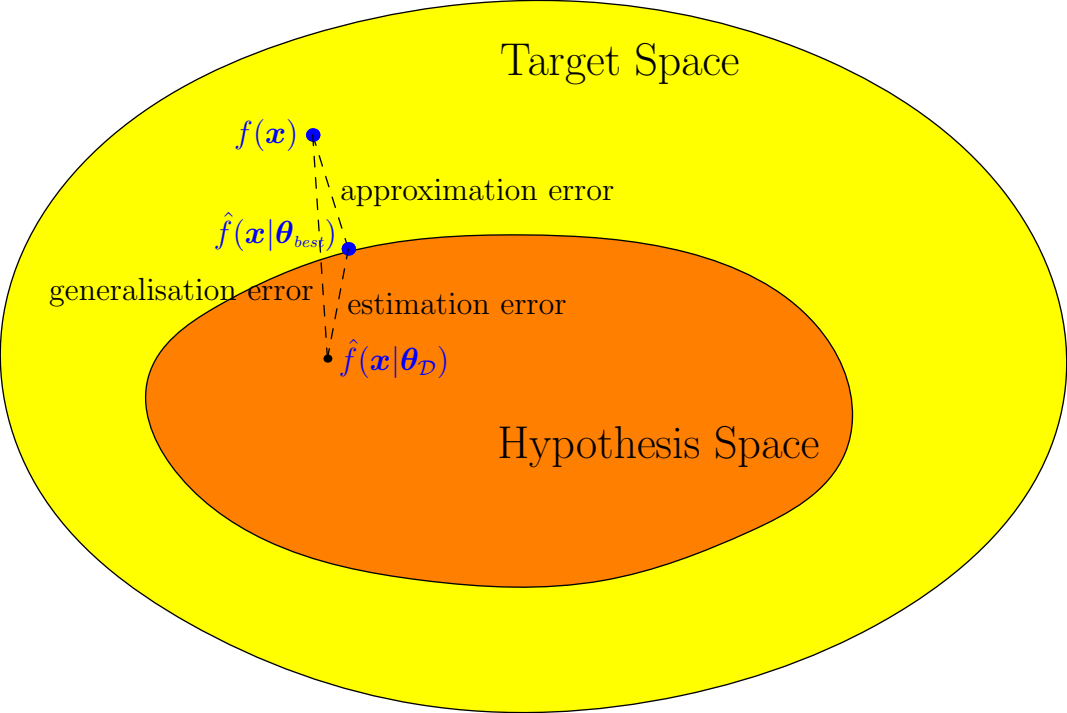

Target Space
$f(\boldsymbol{x})$
Hypothesis Space

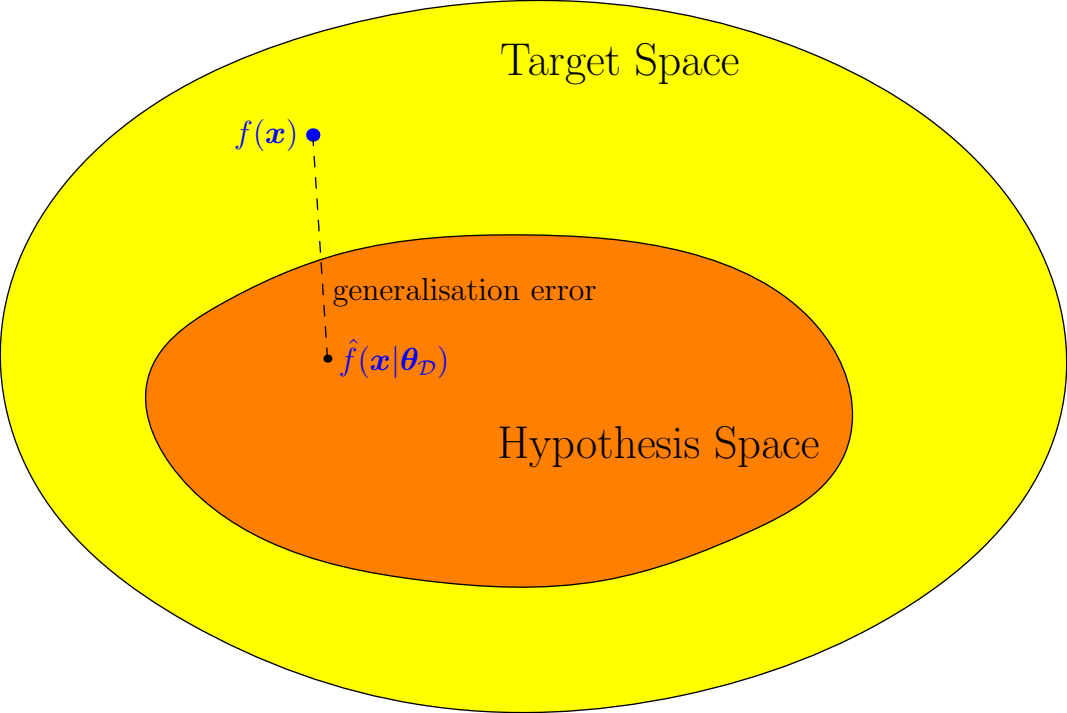

Target Space
$f(\boldsymbol{x})$
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{best})$
Hypothesis Space

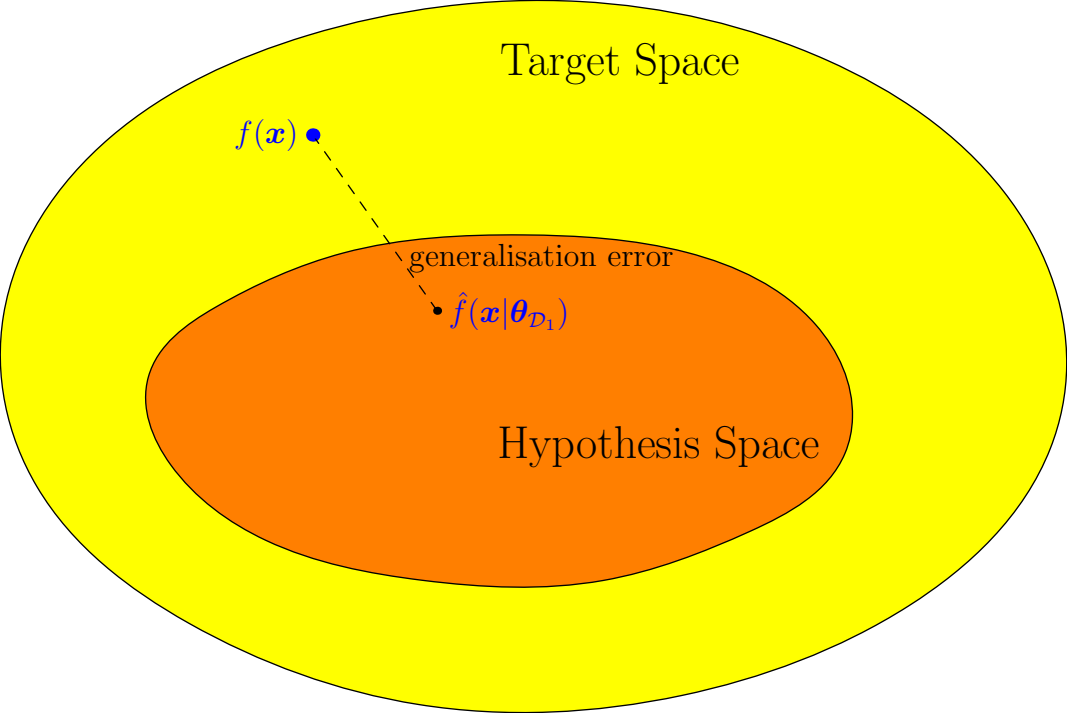

Target Space
$f(\boldsymbol{x})$
approximation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{best})$
Hypothesis Space

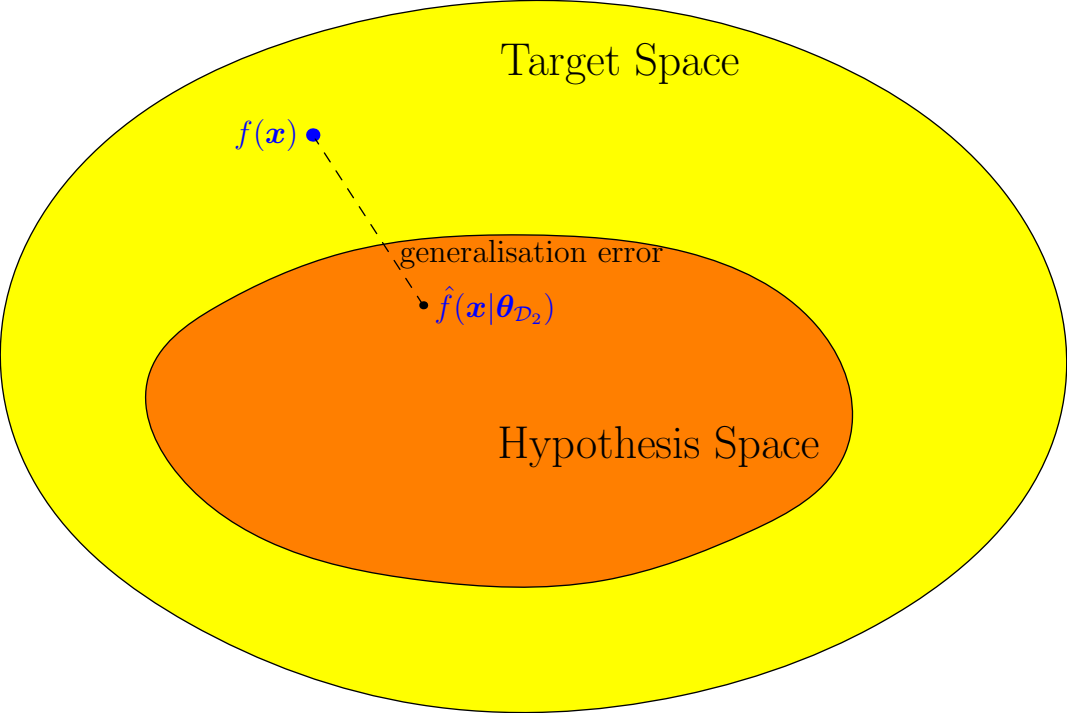

Target Space
$f(\boldsymbol{x})$
approximation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{best})$
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_D)$
Hypothesis Space

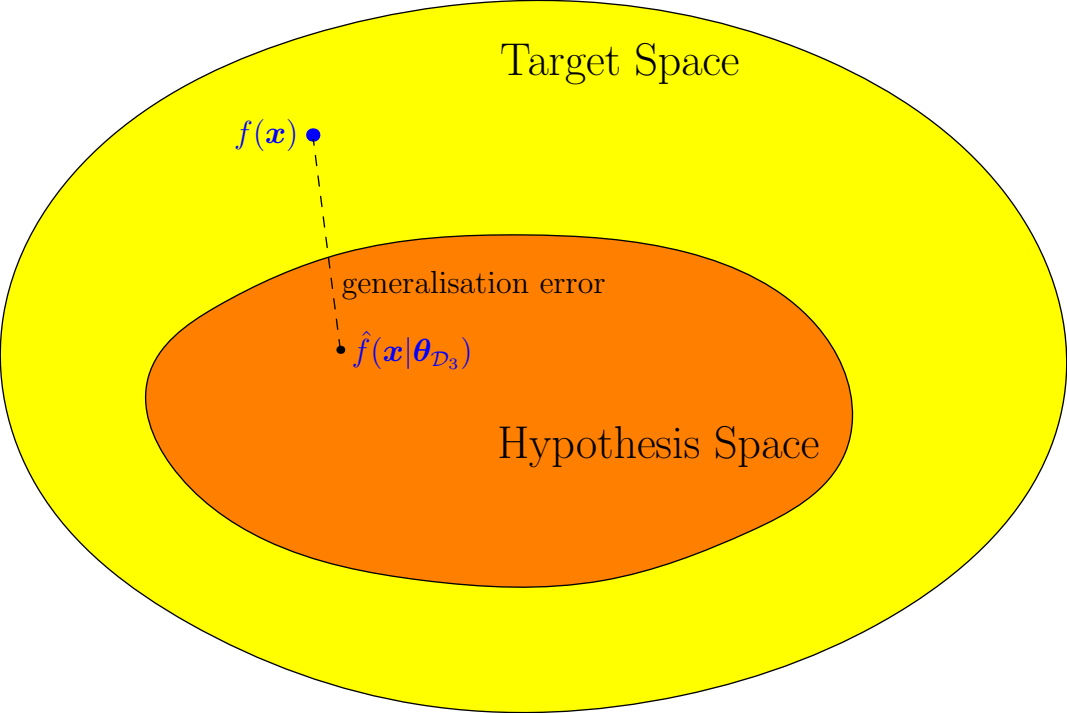

Target Space
$f(\boldsymbol{x})$
approximation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{best})$
generalisation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_D)$
Hypothesis Space

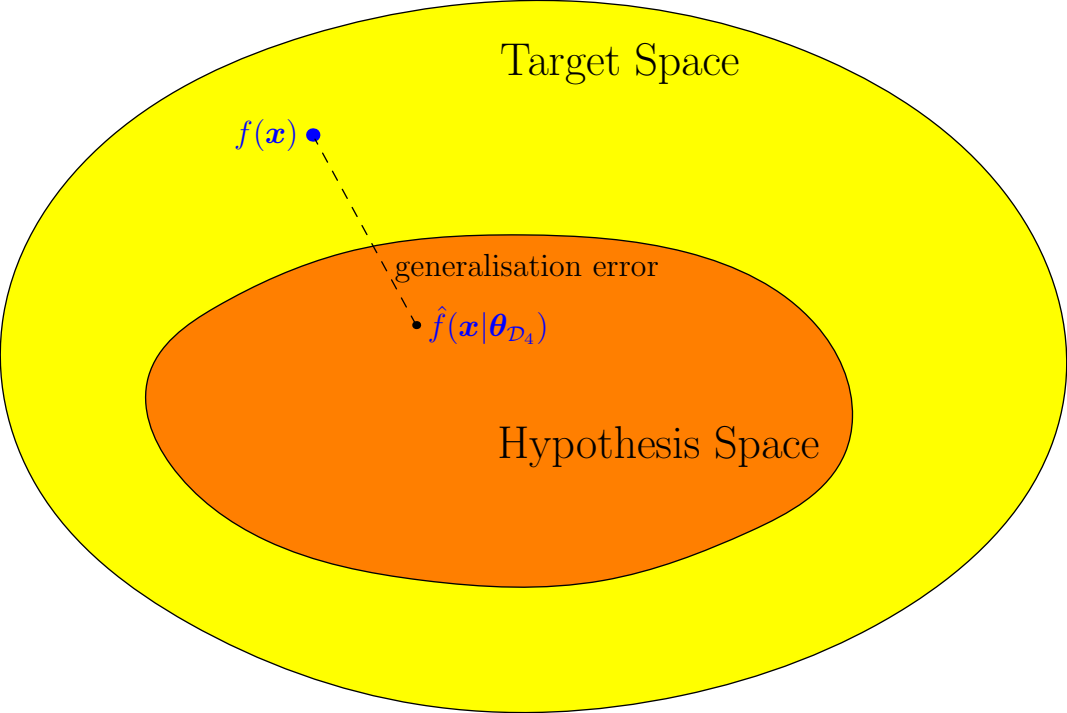

Target Space
$f(\boldsymbol{x})$
approximation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{best})$
generalisation error
estimation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_D)$
Hypothesis Space

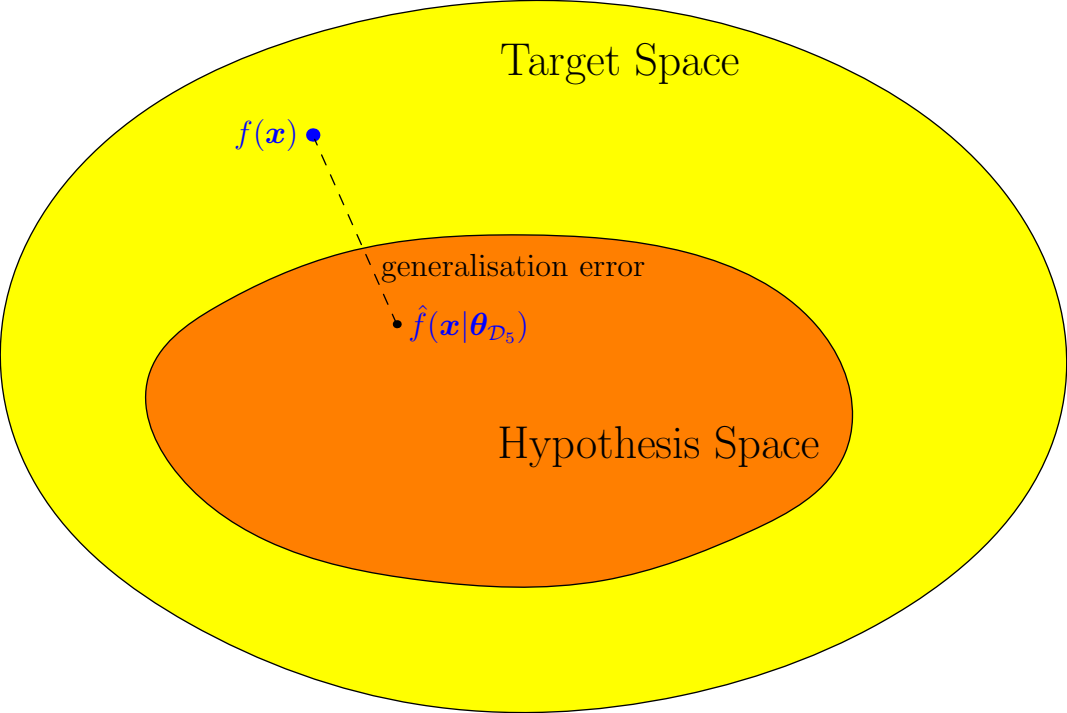

Target Space
$f(\boldsymbol{x})$
generalisation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_D)$
Hypothesis Space

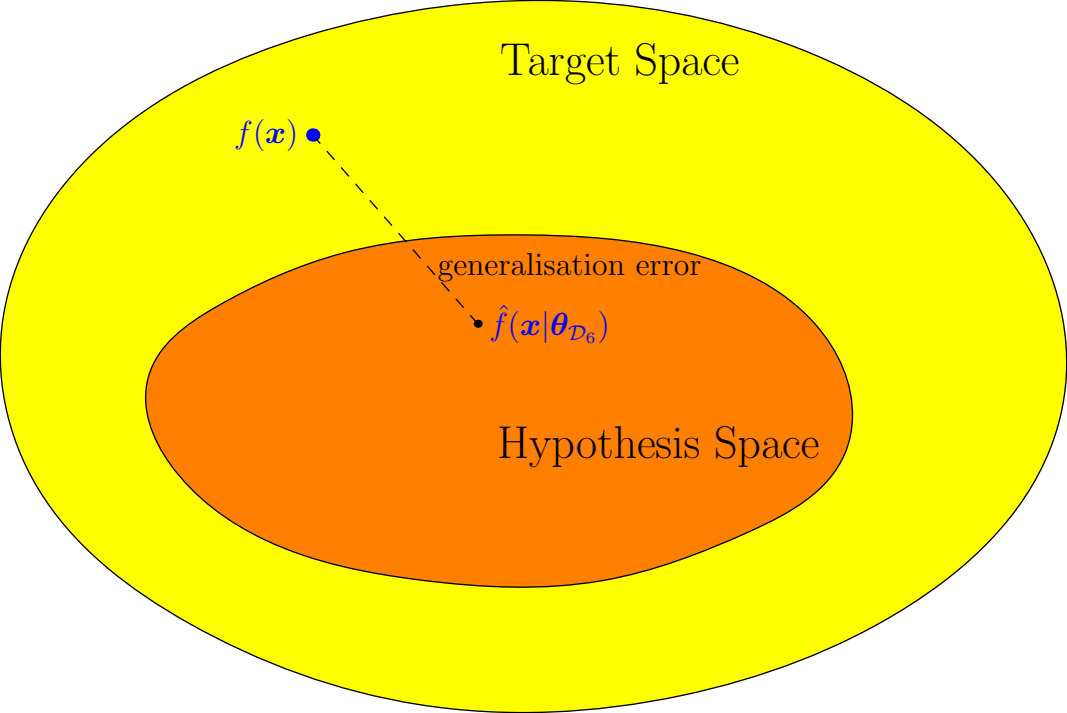

Target Space
$f(\boldsymbol{x})$
generalisation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}_1})$
Hypothesis Space

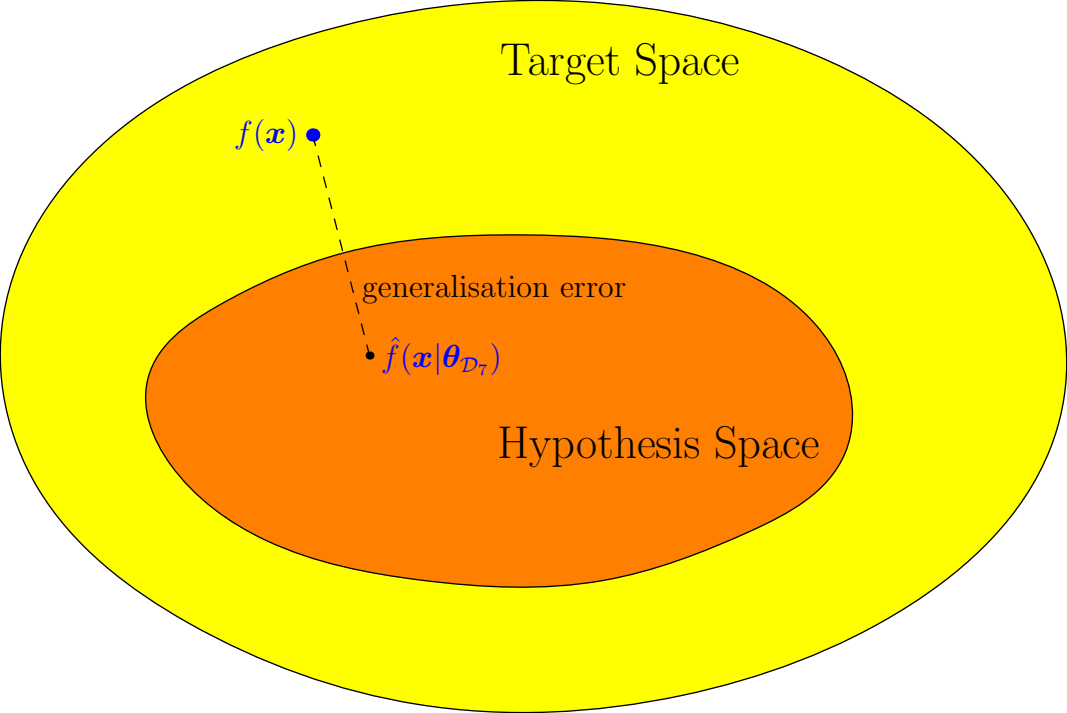

Target Space
$f(\boldsymbol{x})$
generalisation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}_2})$
Hypothesis Space

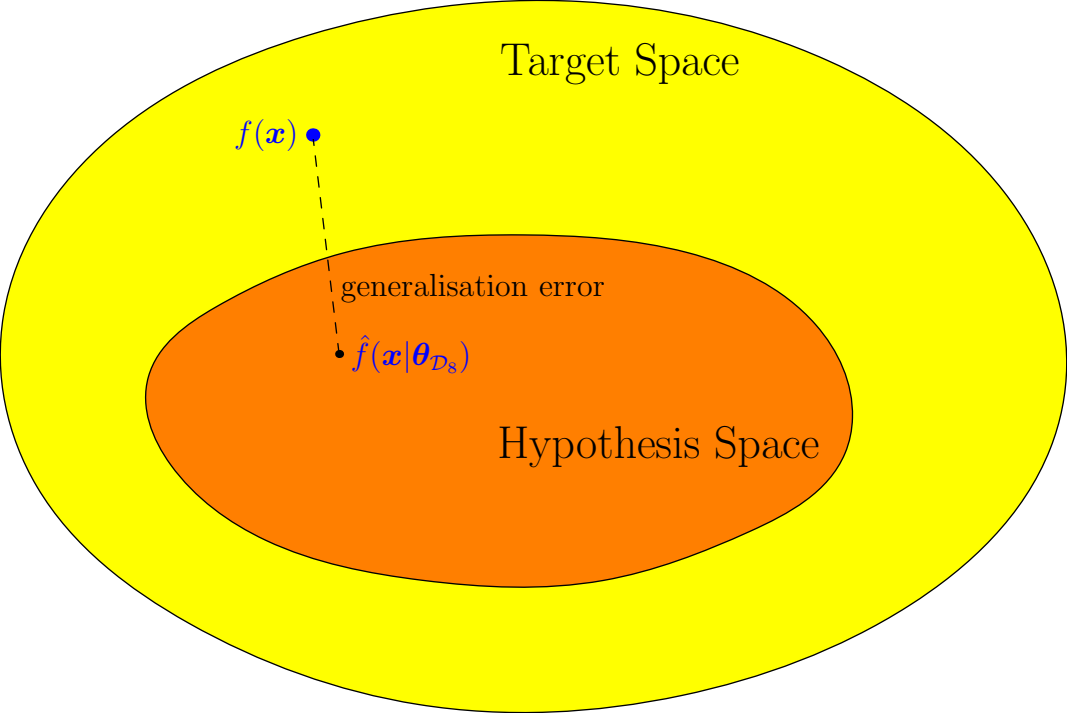

Target Space
$f(\boldsymbol{x})$
generalisation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}_3})$
Hypothesis Space

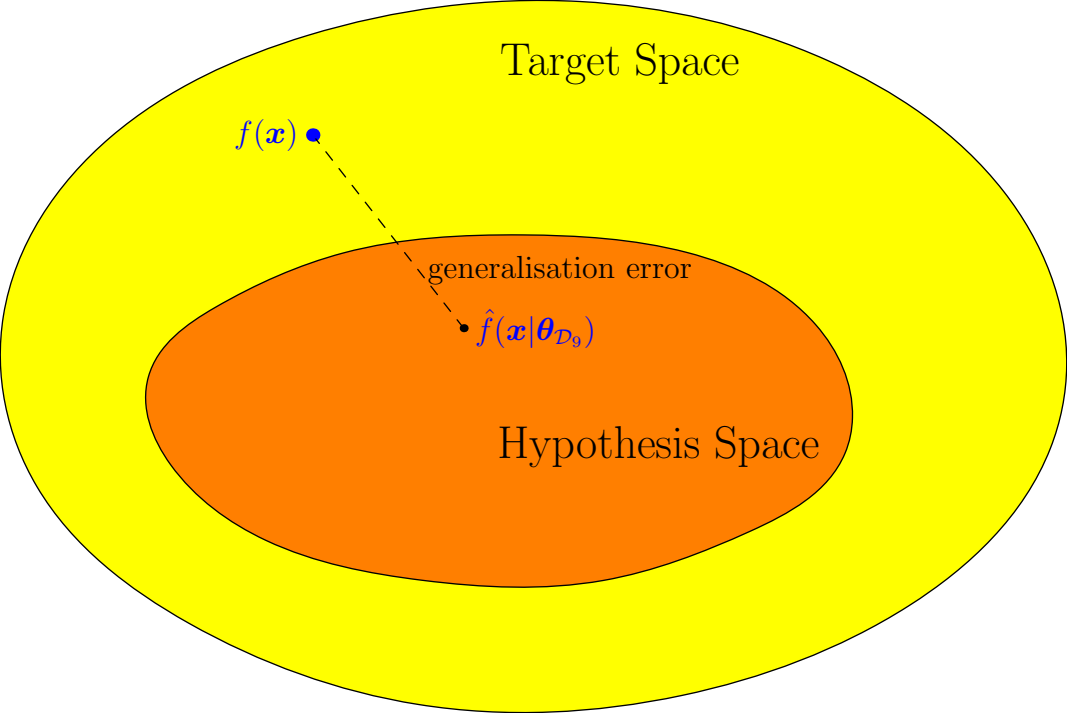

Target Space
$f(\boldsymbol{x})$
generalisation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}_4})$
Hypothesis Space

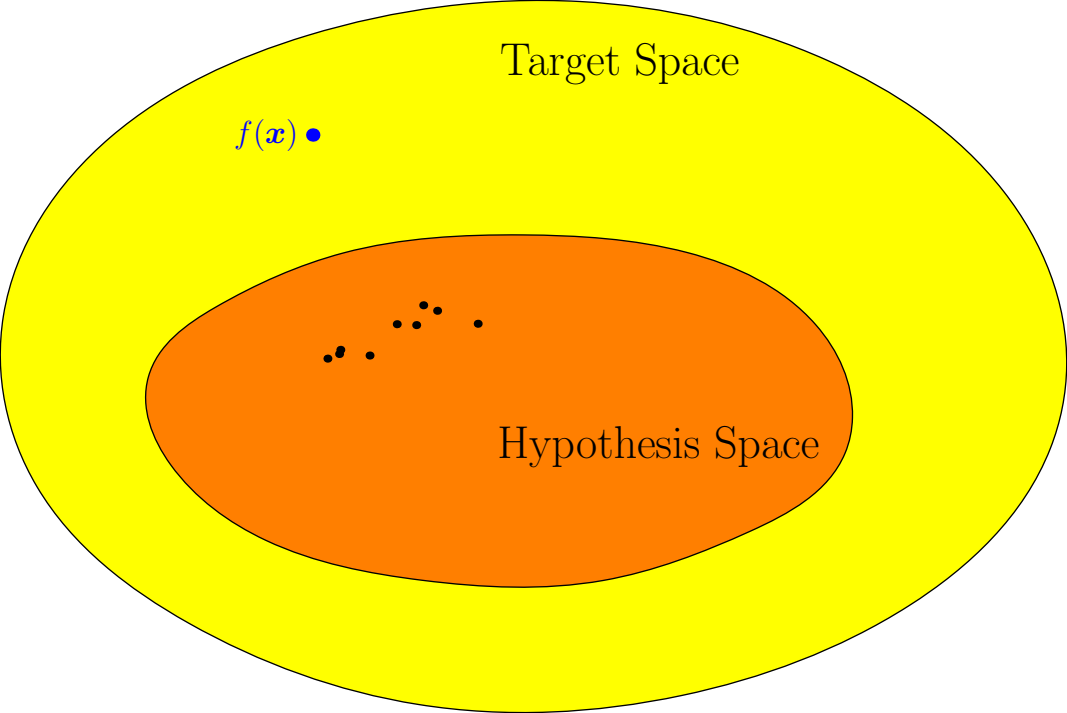

Target Space
$f(\boldsymbol{x})$
generalisation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}_5})$
Hypothesis Space

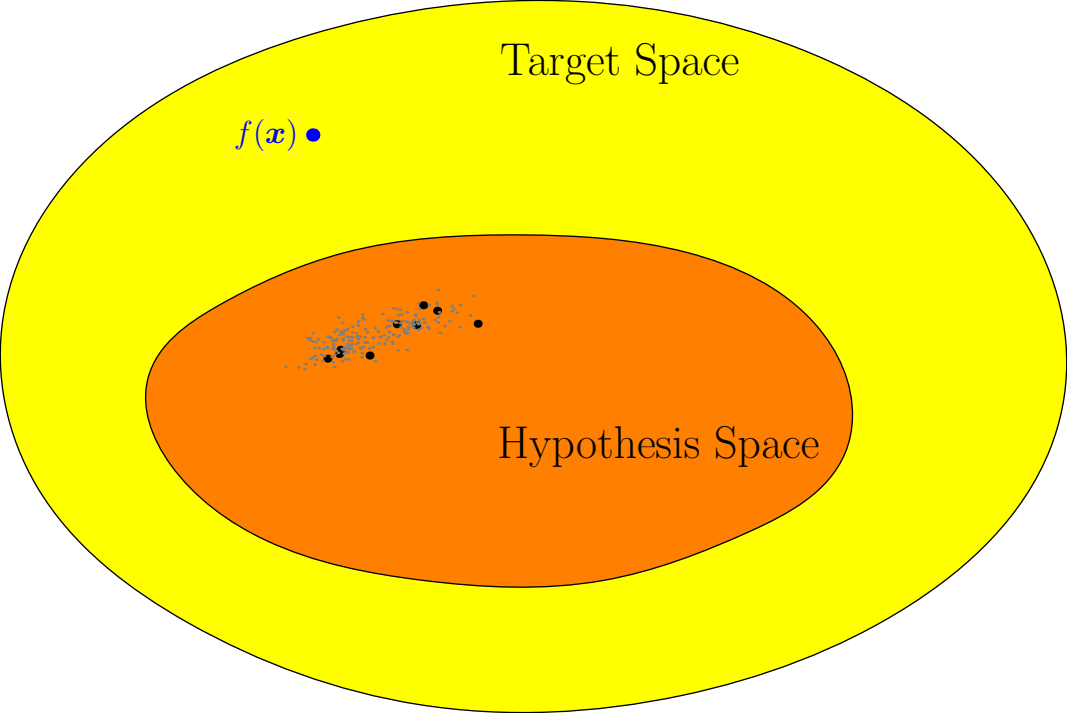

Target Space
$f(\boldsymbol{x})$
generalisation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}_6})$
Hypothesis Space

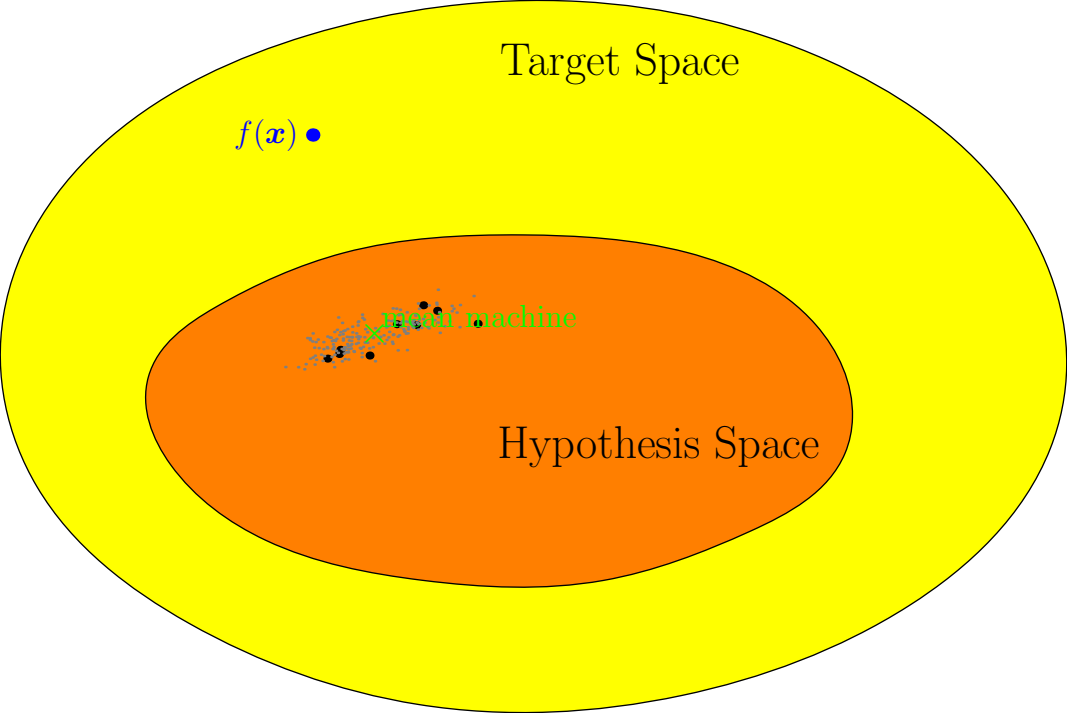

Target Space
$f(\boldsymbol{x})$
generalisation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}_7})$
Hypothesis Space

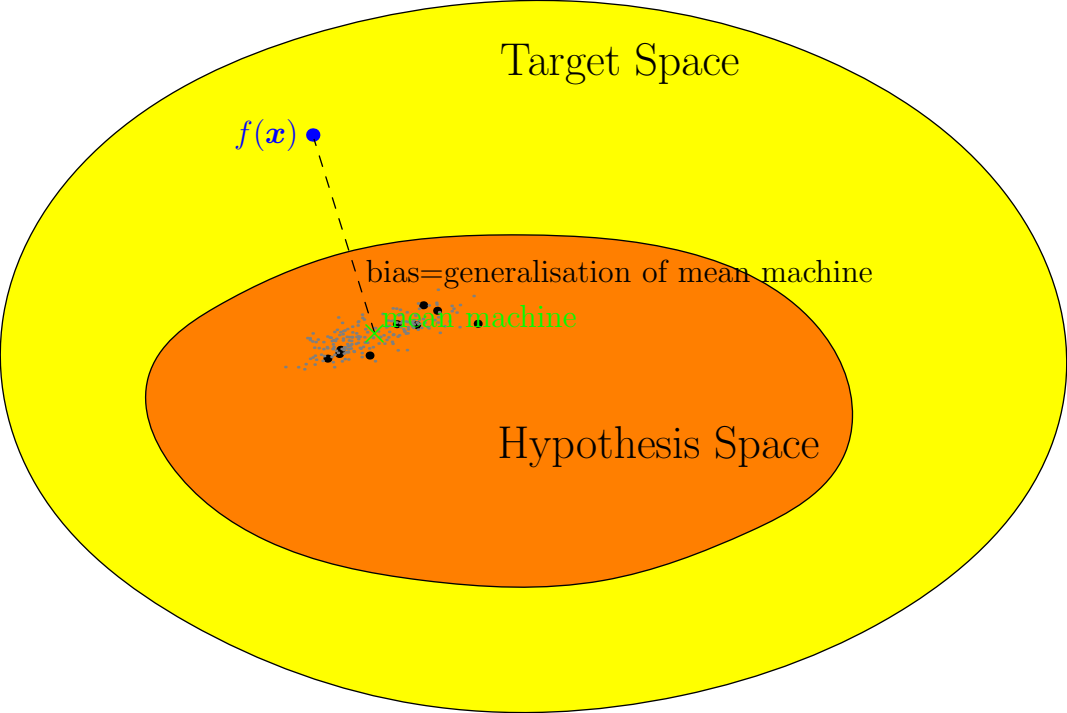

Target Space
$f(\boldsymbol{x})$
generalisation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}_8})$
Hypothesis Space

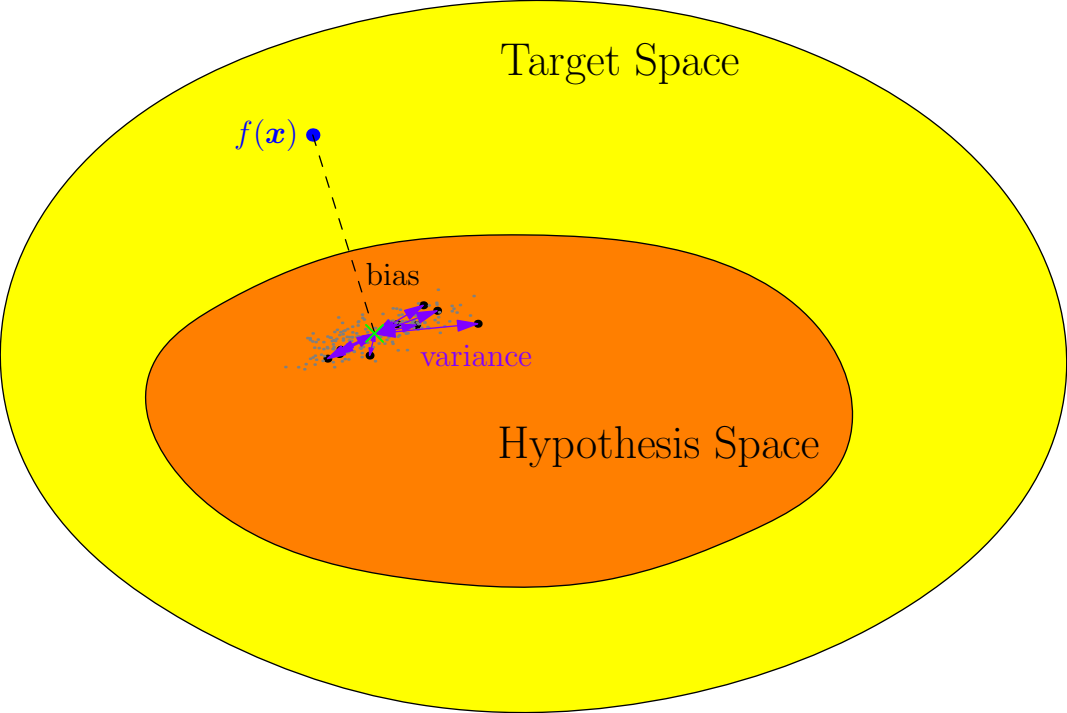

Target Space
$f(\boldsymbol{x})$
generalisation error
$\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}_9})$
Hypothesis Space


Target Space
$f(\boldsymbol{x})$
Hypothesis Space


Target Space
$f(\boldsymbol{x})$
Hypothesis Space


Target Space
$f(\boldsymbol{x})$
learning machine
Hypothesis Space


Target Space
$f(\boldsymbol{x})$
bias=generalisation of mean machine
mean machine
Hypothesis Space


Target Space
$f(\boldsymbol{x})$
bias
variance
Hypothesis Space